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Aim: Sequence Diagram and Collaboration (Communication) Diagram for E-Learning Management System

Implementation:

The Sequence Diagram was created to illustrate the dynamic behaviour of the E-Learning Management System and the sequence in which different system components communicate during various operations. It models important use case flows including course management, student enrollment, accessing course content, assignment submission, quiz attempts, grading and notifications, and live class scheduling.

The main participants involved in these interactions are the Student, Instructor, E-LMS System, Database, Notification Service, Email Service, and Live Class Provider. Each participant is represented using a separate lifeline, while the messages exchanged between them are shown in chronological order from top to bottom. Activation bars indicate the duration for which a participant is actively executing or processing a particular request.

For operations where the system behaviour changes depending on a condition, alt (alternative) combined fragments were used. These fragments represent different possible execution paths, such as whether a course has available seats or is full, whether an assignment is submitted before or after the deadline, whether the quiz timer expires before submission, and whether a student attempts to join a live class within the permitted joining period.

After completing the sequence diagram, the same interactions were represented using a Collaboration (Communication) Diagram. While the sequence diagram mainly focuses on the chronological flow of messages, the collaboration diagram highlights the relationships and communication links between the participating objects. Since it does not use a time axis, the order of communication is represented using numbered messages. Nested

numbering such as 1.1, 2.1, and 6.1–6.3 is used for messages generated as a result of another interaction, making the sequence and dependency between operations clear.

Problem description:

The E-Learning Management System (E-LMS) acts as a centralized platform that allows students and instructors to perform different academic activities associated with online learning. Instructors can create and maintain courses, provide learning resources, assess and grade submissions, record attendance, and organize live classes. Students can enroll in courses, access study materials, submit assignments, take quizzes, check their grades, and participate in scheduled live sessions.

All major user requests are handled by the central E-LMS system, which communicates with other components whenever required. The Database is responsible for storing and retrieving course, student, submission, and session information. The Notification and Email Services are used to inform users about important updates such as the release of grades, while the Live Class Provider handles the creation and access of online class sessions.

The Sequence and Collaboration Diagrams represent these interactions through five main functional areas: Course Management, Enrollment & Access, Assignments & Quizzes, Grading & Notifications, and Live Classes.

Following is a breakdown of the various parts considered while designing the Class Diagram:

[1] Objects / Participants Identified

Based on the different use case interactions, the following users and system components were identified as the main participants:

- Student
- Instructor
- E-LMS (System)
- Database
- Notification Service
- Email Service (SMTP)
- Live Class Provider

[2] Major Use Case Flows Represented

The diagrams model the main activities performed within the E-Learning Management System:

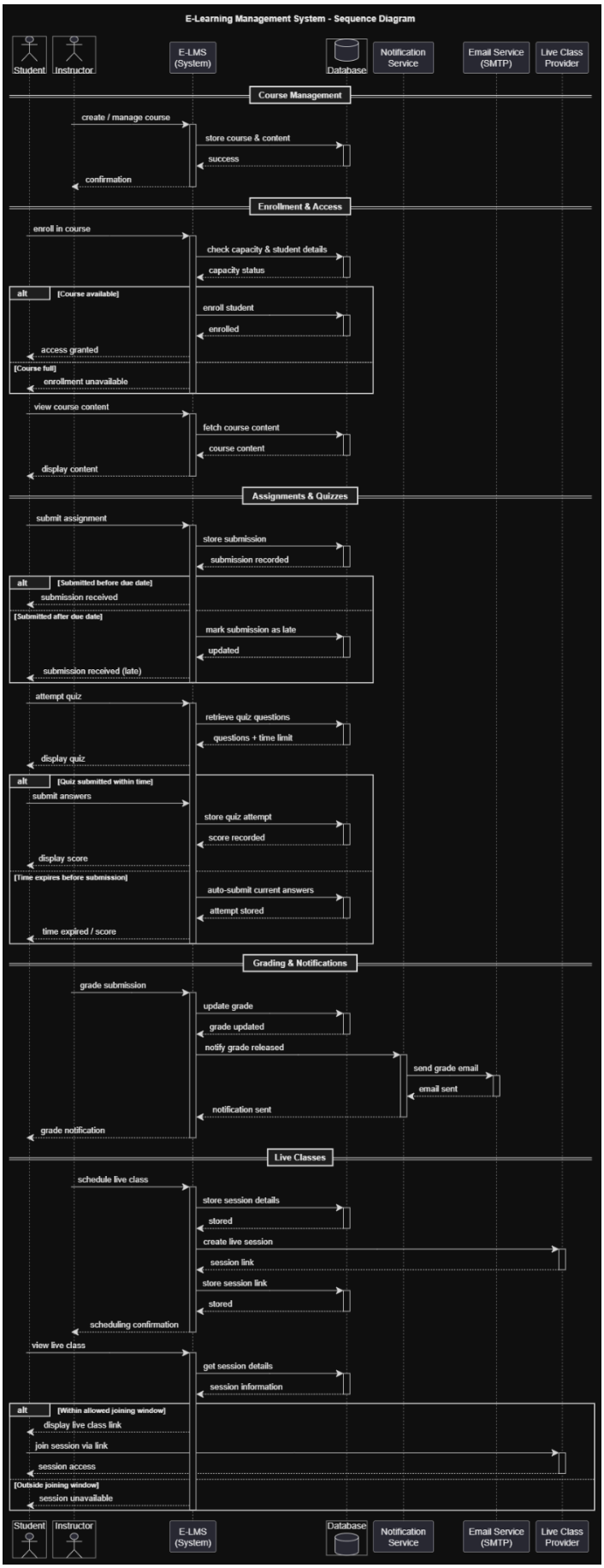
- Course Management – allows instructors to create and manage courses and save the related learning content.
- Enrollment & Access – handles student enrollment, verifies course availability, and provides access to course materials.
- Assignments & Quizzes – covers assignment submissions, quiz attempts, and storing the corresponding submission details.
- Grading & Notifications – manages the evaluation of submissions and sends grade-related notifications to students through email.
- Live Classes – supports scheduling online sessions and enables students to access and join the scheduled classes.

[3] Conditional Logic Using `alt` Fragments

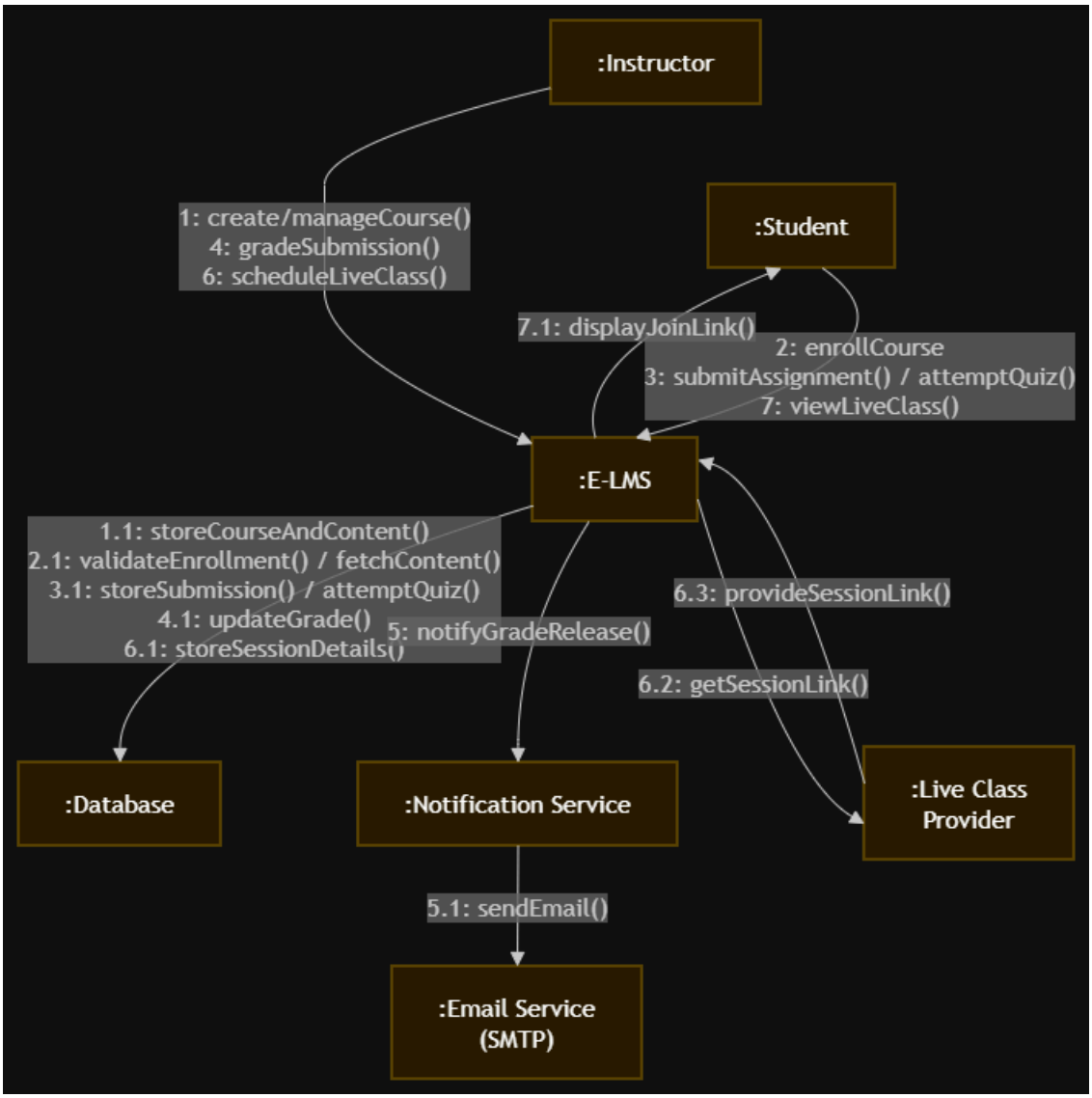
Alternative (`alt`) fragments were included in the Sequence Diagram to represent situations where the system can follow different paths depending on a condition:

- [Course available] / [Course full] – determines whether a student can successfully enroll in a course.
- [Submitted before due date] / [Submitted after due date] – distinguishes between an on-time and a late assignment submission.
- [Time expires before submission] – represents the case where the quiz duration ends before the student submits the answers.
- [Within allowed joining window] / [Outside joining window] – checks whether the student is permitted to enter the scheduled live class.

Sequence diagram:


E-Learning Management System - Sequence Diagram
Student
Instructor
E-LMS (System)
Database
Notification Service
Email Service (SMTP)
Live Class Provider
Course Management
create / manage course
store course & content
success
confirmation
Enrollment & Access
enroll in course
check capacity & student details
capacity status
alt
[Course available]
enroll student
enrolled
access granted
[Course full]
enrollment unavailable
view course content
fetch course content
course content
display content
Assignments & Quizzes
submit assignment
store submission
submission recorded
alt
[Submitted before due date]
submission received
[Submitted after due date]
mark submission as late
updated
submission received (late)
attempt quiz
retrieve quiz questions
questions + time limit
display quiz
alt
[Quiz submitted within time]
submit answers
store quiz attempt
score recorded
display score
[Time expires before submission]
auto-submit current answers
attempt stored
time expired / score
Grading & Notifications
grade submission
update grade
grade updated
notify grade released
send grade email
email sent
notification sent
grade notification
Live Classes
schedule live class
store session details
stored
create live session
session link
store session link
stored
scheduling confirmation
view live class
get session details
session information
alt
[Within allowed joining window]
display live class link
join session via link
session access
[Outside joining window]
session unavailable

Collaboration diagram:



The interactions shown in the Sequence Diagram are represented in the Collaboration Diagram as numbered communication links between the participating objects. The main messages, including 1: create/manageCourse(), 2: enrollCourse(), 3: submitAssignment()/attemptQuiz(), 4: gradeSubmission(), 5: notifyGradeRelease(), 6: scheduleLiveClass(), and 7: viewLiveClass(), are initiated by either the Instructor or Student and sent to the E-LMS system.

Whenever a main interaction requires additional processing, it is represented using nested message numbers. For instance, message 1 leads to 1.1: storeCourseAndContent(), which communicates with the Database. Similarly, scheduling a live class through message 6 results in 6.1: storeSessionDetails(), followed by 6.2: getSessionLink() and 6.3: provideSessionLink() from the Live Class Provider. For message 7, the system responds to the Student through 7.1: displayJoinLink().

Thus, the use of decimal sequence numbering clearly indicates the order and dependency of interactions between objects, even though a Collaboration Diagram does not represent communication along a chronological time axis.

Conclusion:
The Sequence and Collaboration Diagrams show how the different components of the E-Learning Management System interact with each other. The Sequence Diagram mainly focuses on the order in which messages are exchanged and uses `alt` fragments to represent conditions like course availability, late submissions, quiz time limits, and live-class access. On the other hand, the Collaboration Diagram focuses more on how the different objects are connected and communicate, using numbered messages to show their order. Overall, this experiment helped in understanding how the same E-LMS workflow can be represented in two different UML diagrams, with each providing a different view of the system's behaviour.